A Foundation Model for Lesion Segmentation on Brain MRI With Mixture of Modality Experts

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***Abstract*— Brain lesion segmentation is crucial for neurological disease research and diagnosis. As different types of lesions exhibit distinct characteristics on different imaging modalities, segmentation methods are typically developed in a task-specific manner, where each segmentation model is tailored to a specific lesion type and modality. However, the use of task-specific models requires predetermination of the lesion type and imaging modality, which complicates their deployment in real-world scenarios. In this work, we propose a universal foundation model for brain lesion segmentation on *magnetic resonance imaging* (MRI), which can automatically segment different types of brain lesions given input of various MRI modalities. We develop a novel *Mixture of Modality Experts* (MoME) framework with multiple expert networks attending to different imaging modalities. A hierarchical gating network is proposed to combine the expert predictions and foster expertise collaboration. Moreover, to avoid the degeneration of each expert network, we introduce a curriculum learning strategy during training to preserve the specialisation of each expert. In addition to MoME, to handle the combination of multiple input modalities, we propose MoME+, which uses a soft dispatch network for input modality routing. We evaluated the proposed method on nine brain lesion datasets, encompassing five imaging modalities and eight lesion types. The results show that our model outperforms state-of-the-art universal models for brain lesion segmentation and achieves promising generalisation performance onto unseen datasets.**

***Index Terms*— Foundation model, mixture of experts, brain imaging, brain lesion segmentation.**

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I. Introduction

BRAIN lesion segmentation plays an essential role in neurological research and diagnosis by offering a quantitative description of pathologies [1], [2]. Deep learning has greatly advanced the progress of brain lesion segmentation in recent years [3], [4], [5], [6], [7], [8]. Since brain lesions tend to manifest with different characteristics due to various patient-dependent pathological alterations, the segmentation of different types of brain lesions requires the use of different *magnetic resonance imaging* (MRI) modalities [9], [10], [11], such as T1-weighted (T1w), T2-weighted (T2w), T1 contrast-enhanced (T1ce), FLAIR, diffusion weighted imaging (DWI), etc. They are either used standalone or as a combination [12], [13], [14], which increases the complexity of segmentation model development. In the traditional paradigm (Fig. 1a), task-specific models are developed, where each model segments brain lesions on a specific modality or modality combination [2], [15]. In addition, for different types of brain lesions, such as stroke lesions, white-matter hyperintensities (WMH), glioma, etc., different task-specific models need to be developed. At the deployment stage, among all these task-specific models, clinicians are compelled to manually choose the optimal model for each patient MRI scan. This process hinders the practical deployment of *artificial intelligence* (AI) models. Contrary to this traditional paradigm, we propose a novel task-agnostic universal foundational model (Fig. 1b), which is capable of automatically segmenting various types of brain lesions regardless of the input MRI modality. The model can also be extended to handle heterogeneous inputs with multiple modalities.

Foundation models have gained popularity in medical image segmentation, thanks to the success of the *Segment Anything Model* (SAM) [16] and its variants [17], [18], [19], [20], as well as other innovative foundation models specifically developed for medical imaging [21], [22], [23], [24]. Earlier

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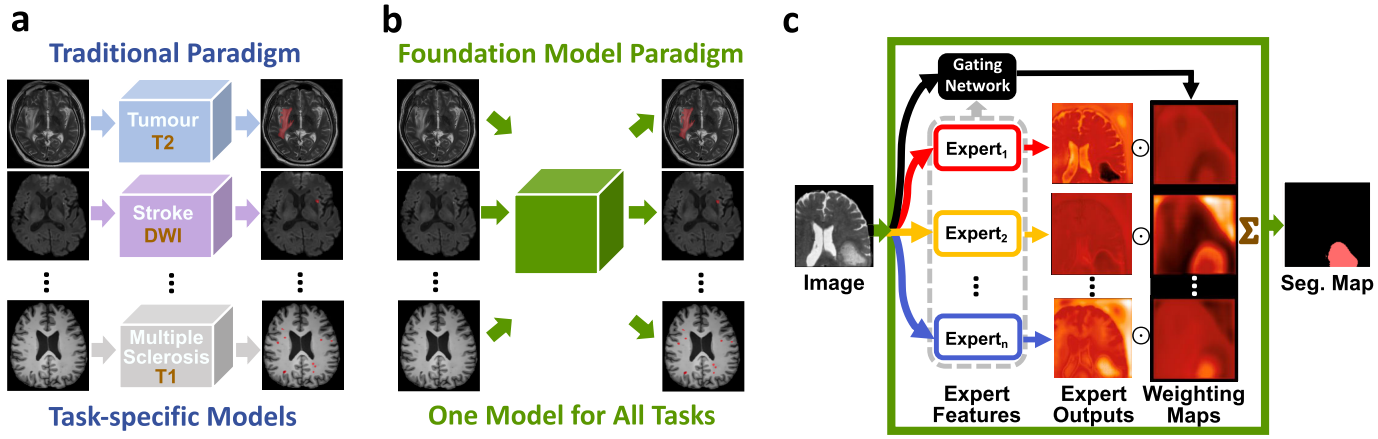


Fig. 1. Different paradigms for brain lesion segmentation and an overview of the proposed framework: a) the traditional paradigm that trains multiple task-specific models; b) the foundation model paradigm that trains a single universal model for multiple tasks; c) the proposed *mixture of modality experts* (MoME) model. In MoME, the input image passes through each modality expert to generate individual expert output. The gating network takes the image and the expert feature maps as input. It then generates voxel-wise weighting maps, which aggregate the expert outputs to obtain the final segmentation map.

works aim to develop a universal model across all modalities, organs, and tasks. However, due to the complexity of medical imaging data and tasks, the segmentation accuracy of such a universal model is often limited [25]. Thus, foundation models focusing on narrower scopes are explored, which are designed for a particular modality or a particular organ/task [25]. For example, several models are proposed specifically for *computed tomography* (CT) image segmentation [22], [26], [27], showing promising results for anatomical structure and large tumour segmentation. Despite the progress, most existing foundation models may not be directly applicable to brain lesion segmentation [17], [23]. Most noticeably, universal brain lesion segmentation requires the capability of handling the diversity of input imaging modalities, which is not addressed by many of the existing foundation models.

In this study, inspired by *Mixture of Experts* (MoE) [28], [29], [30], we propose a *Mixture of Modality Experts* (MoME) foundational model for brain lesion segmentation on MRI, illustrated in Fig. 1c. Our main contributions are summarised as follows:

- MoME is developed to be task-agnostic and capable of handling diverse imaging modalities and accommodating various types of brain lesions. Each expert specialises in processing certain image modalities.
- To explore collaboration among the experts, a hierarchical gating network is designed to combine expert predictions and facilitate collaborative expertise exploration layer by layer. Additionally, to keep the specialisability of each expert, an anti-degeneration strategy based on curriculum learning is developed.
- We conducted comprehensive experiments on nine brain imaging datasets comprising a total number of 6,585 annotated 3D images, with five MRI modalities and eight lesion types. The results show that MoME outperforms competing foundation models in multiple aspects and provides improved generalisation onto unseen datasets.

A preliminary version of this work has been published in MICCAI 2024 [31], focusing on using a single image as input. Here, we have substantially extended the conference version. The MoME model has been extended to MoME+ to handle combined multiple modalities as input. To this end, we develop a novel dispatch network with a soft assignment strategy and prior constraints, and employ random dropout of modalities during training to mimic the heterogeneous input. When a maximum of N modalities are available, a single unified MoME+ achieves segmentation performance comparable to that of $2^N - 1$ dedicated nnU-Nets for different modality combinations. Additionally, we include new experimental results for handling missing modalities, investigate strategies of decoder sharing for multiple experts to improve model efficiency, as well as extend the illustration and discussion.

The proposed MoME and MoME+ models demonstrate promising segmentation performance that is agnostic against modalities and modality combinations. Since only a single universal model is needed, it eases model management and deployment, making it applicable to a broader range of scenarios. The code of MoME and MoME+ is available at https://github.com/ZhangxinruBIT/MoME.

II. Related Work

A. Brain Lesion Segmentation and Medical Imaging Foundation Models

The task of brain lesion segmentation differs significantly from natural image segmentation or other anatomical structure segmentation tasks. Unlike anatomical structure segmentation, brain lesion segmentation deals with a broader variations of data distributions, in terms of different lesion types, varying spatial locations, scattering of multiple lesions, and irregular shapes [1], [32], which poses substantial challenges for developing segmentation foundation models.

Reference [23] proposed a universal medical image segmentation model, Hermes, which allows modality information and target segmented objects to be embedded as context priors into the model. However, brain lesion types and their locations are often patient-specific, which are difficult to be embedded as context priors. SAM-inspired models such as SAM-Med3D [17] and MedSAM [19] have demonstrated the

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potential of foundational segmentation models for 3D medical image segmentation. These models require input prompts in the format of click points or bounding boxes. While prompt generation is relatively easy for a single large tumour that appears as a central cluster in the brain, e.g., glioma, it can be more challenging to define for other types of lesions, e.g., stroke lesions that manifest as small and challenging-to-detect anomalies (see Fig. 1a) [2]. Additionally, *white matter hyperintensity* (WMH) are often widely dispersed [5], [33] and scattered across the brain at multiple locations (see Fig. 1a), which can be easily missed by point-based or bounding box-based prompts. Neuralizer [4] proposed a different paradigm for developing medical imaging foundational models by leveraging a context set. After the model pre-training stage, the model generalises to a new task by learning transferable features from a context set that contains paired images and segmentation masks, similar to few-shot learning. Neuralizer is mainly evaluated on 2D coronal slices instead of 3D volumes, possibly due to the computational resource limitation in learning from a 3D context set.

B. Mixture of Experts

The mixture of experts (MoE) approach is widely acknowledged for its efficacy in scaling up foundation models and improving their performance [34], [35], [36], particularly in handling larger models and datasets with strong generalisation capabilities [37], [38], [39]. MoE involves creating multiple distinct experts tasked with learning both shared and task-specific knowledge, while employing a gating function to regulate the contribution of each expert. MoE has been explored for applications in *natural language processing* (NLP) [40], [41], multi-modal learning [30], [42], and computer vision [28], [43], [44], [45]. In computer vision, MoE has been demonstrated effective in image classification [28], [43] and segmentation tasks [28], [44]. In the field of medical image analysis, Patcher [28] regarded the feature maps extracted from different resolution levels as expert outputs and introduced a MoE decoder to integrate multi-level global-to-fine features. Here, we propose that the definition of experts in the MoE model could be carefully tailored to specific tasks and data domains. Brain lesion segmentation involves multiple imaging modalities. Several studies have shown that utilising the modality information benefits the segmentation performance [46], [47]. There is potential to leverage the characteristics of each modality to define the expert for each modality under the MoE framework.

C. Heterogeneous Input With Missing Modalities

In the field of medical imaging, the input data are inherently heterogeneous [48], with various imaging modalities or combination of modalities. A common issue that requires attention is the modality missing problem, where one or several of the multiple imaging modalities are missing [12], [13], [14], [46], [47], [49]. To address this issue, strategies such as synthesizing missing modalities using a generative model [50] or learning a modality-invariant feature space [46], [51], [52] have been explored. If N imaging modalities are needed for a segmentation task and some modalities are missing, the total number of combinations of remaining available modalities is $2^N - 1$. Consequently, numerous input-specific segmentation models need to be trained, dedicated to each modality combination [14], [53]. Wang et al. proposed non-dedicated training, which results in N models and reduces the model complexity [12]. The non-dedicated training approach, while requiring fewer models for training, may compromise performance to some extent compared to dedicated training [12]. In this work, we explore a different approach, which is to develop a single foundation model capable of handling different modality combinations. Such a foundation model would offer convenience and ease for model development and deployment.

III. Methods

A. Method Overview

The architecture of MoME features two major components, namely the expert networks and the gating network. Each expert network is specialised in a specific domain of imaging modality. The gating network acts as a decision-maker, ensembling the expert network outputs and dynamically adjusting the ensemble weights with hierarchical guidance. While training the segmentation model, an anti-degeneration strategy based on curriculum learning is employed, which enables the model training to gradually transition from specialising the role of each expert to determining the entire model through collaboration. We develop our method for two types of input data, only a single but any modality as input (MoME, Sec. III-B) and a combination of multiple modalities as input (MoME+, Sec. III-C).

B. MoME for Single Input Image

1) Expert Specialisation: Suppose there are N imaging modalities $\{\mathcal{M}_i\}_{i=1}^{N}$ of interest, where $\mathcal{M}_i$ represents the i-th modality. We aim to develop a modality-agnostic model that can perform segmentation for any single input image, regardless of its imaging modality. We focus on the five most widely used brain MRI modalities, including T1w, T2w, T1ce, FLAIR, and DWI, and thus $N = 5$. To address the problem of modality diversity, in MoME, N expert networks $\{E_i\}_{i=1}^{N}$ are developed, where E_i denotes the i-th expert. Each expert E_i is expected to specialise in a specific modality $\mathcal{M}_i$. The single input image will go through all expert networks, with their outputs ensembled by the gating network to form the final output. Without losing generality, we assume that each expert network follows a multi-resolution encoder-decoder architecture, which is common for medical image segmentation such as in U-Net [54]. Note that the experts here are built for 3D brain image segmentation, distinguishing them from sparse MoE architectures for 2D natural images where a higher number of experts can be afforded [55], [56].

2) Hierarchical Gating Network: Following the specialisation of expert networks, we propose a gating network G that facilitates the collaboration and conducts voxel-wise weighted ensemble among the expert outputs. Note that different imaging modalities can be correlated while providing complementary information. For example, FLAIR appears similar

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to T2w but with fluid being attenuated; for an input T2w image, the outputs of the FLAIR expert and T2w expert may provide complementary information that both contribute to lesion segmentation. Thus, to enhance collaboration between experts, the gating network fuses the multi-resolution outputs from multiple experts instead of selecting one of them.

Mathematically, given an input image $\mathbf{X}$, each expert E_i generates outputs at multiple resolution levels on the decoder side, denoted by $E_i(\mathbf{X}) = \{\mathbf{e}_i^l\}_{l=1}^L$, where $\mathbf{e}_i^l$ is the output of E_i at the l-th resolution level and L indicates the number of resolution levels. The gating network G generates a set of hierarchical voxel-wise weight maps at different resolution levels for each expert, denoted by $\{\mathbf{g}_i^l\}_{l=1}^L$. Combining the expert networks and the gating network, the final output $\mathbf{O}$ is formulated as

$$\mathbf{O} = \left\{\mathbf{o}^l\right\}_{l=1}^L \quad \text{with} \quad \mathbf{o}^l = \sum_{i=1}^N \mathbf{e}_i^l \odot \mathbf{g}_i^l, \tag{1}$$

where $\mathbf{o}^l$ is the aggregated output of all experts at the l-th resolution level and $\odot$ represents voxel-wise multiplication. The segmentation at the l-th level is obtained by applying the softmax operator to $\mathbf{o}^l$. During training, deep supervision [54], [57] is employed to guide the segmentation at multiple resolution levels. At the inference stage, only the highest resolution level determines the final segmentation result [54], [57].

3) Curriculum Learning to Alleviate Expert Degeneration: Although intuitively it is possible to jointly train the experts and gating network, this may lead to expert degeneration, where each expert no longer focuses on a specific modality and their activation becomes imbalanced [34], [58], reducing the capability of the whole model to handle diverse modalities. To address this issue, we propose a curriculum learning strategy for model training, which gradually transitions from specialising each individual expert to tuning the whole model to encourage expert collaboration and refinement.

The curriculum learning is achieved with a dynamic loss $\mathcal{L}_{\text{cl}}$ that comprises the specialisation loss $\mathcal{L}_i$ for each modality expert E_i and the collaboration loss $\mathcal{L}_{\text{MoME}}$. During training, $\mathcal{L}_{\text{cl}}$ is gradually transitioned from $\mathcal{L}_i$ to $\mathcal{L}_{\text{MoME}}$, enabling dynamic attention, which is controlled by the weight f_{epoch}:

$$\mathcal{L}_{\text{cl}} = f_{\text{epoch}} \cdot \sum_{i=1}^N \mathbb{1}_{\mathbf{X}\in\mathcal{M}_i} \cdot \mathcal{L}_i + (1 - f_{\text{epoch}}) \cdot \mathcal{L}_{\text{MoME}}. \tag{2}$$

Here, $\mathbb{1}_{\mathbf{X}\in\mathcal{M}_i}$ is an indicator function that is one when the input image $\mathbf{X}$ belongs to the modality $\mathcal{M}_i$ and zero otherwise, and f_{epoch} is a dynamic weight that evolves with the epoch iteration. f_{epoch} is designed as $f_{\text{epoch}} = (1 - \text{epoch}_{\text{current}}/\text{epoch}_{\text{total}})^2$, where $\text{epoch}_{\text{current}}$ and $\text{epoch}_{\text{total}}$ are the current epoch and total number of epochs, respectively. In this way, during the early training stage, f_{epoch} is greater, which emphasises the loss $\mathcal{L}_i$ for the expert E_i that matches the modality of the input image $\mathbf{X}$. During the later training stage, f_{epoch} decreases and $\mathcal{L}_{\text{MoME}}$ becomes gradually more emphasised.

$\mathcal{L}_i$ is designed with deep supervision at each resolution level as

$$\mathcal{L}_i = \sum_{l=1}^L k^l \cdot \mathcal{L}(\sigma(\mathbf{e}_i^l), d^l(\mathbf{Y})), \tag{3}$$

where k_l is a weight for the l-th level, $\sigma(\cdot)$ is the softmax activation, $d^l(\cdot)$ is a down-sampling operator that matches the resolution of the ground truth annotation $\mathbf{Y}$ with the expert network prediction at the l-th level, and $\mathcal{L}(\cdot,\cdot)$ is the combination of the soft Dice loss and cross-entropy loss [54]. $\mathcal{L}_{\text{MoME}}$ is formulated with the collaboration between experts as

$$\mathcal{L}_{\text{MoME}} = \sum_{l=1}^L k^l \cdot \mathcal{L}(\sigma(\mathbf{o}^l), d^l(\mathbf{Y})), \tag{4}$$

which encourages the ensembled output $\mathbf{o}^l$ from multiple experts to be close to the ground truth segmentation $\mathbf{Y}$.

C. MoME+: Extended MoME for Multiple Input Images

The previously described MoME model accepts a single input image, regardless of its modality. The single input image goes through multiple experts and their predictions are ensembled. Here we extend its capability to handle multi-modality input images, which are common in brain lesion segmentation, and the number of modalities and their combinations can vary. We refer to this extended model as MoME+ hereinafter. In MoME+, we still have N expert networks, with N denoting the maximum number of modalities. There are multiple input images, the number of which can be smaller than N due to missing modalities. To address the mismatch between the number of available input images and the number of expert networks, and to leverage information from available modalities to infer channels for missing modalities, we develop an additional dispatch network D, which generates a $N \times N$ dispatch matrix, and then dispatches the input images to the N expert networks.

We denote the modality availability information as $\boldsymbol{c} = (c_1, c_2, \ldots, c_N)$, where $c_i = 1$ indicates that the corresponding modality $\mathcal{M}_i$ is available and $c_i = 0$ indicates that it is not. The dispatch network D is a trainable network, which takes $\boldsymbol{c}$ as input and predicts a dispatch matrix $\mathbf{D} \in \mathbb{R}^{N\times N}$. The multi-modality input images are denoted as $\widetilde{\mathbf{X}} = (\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_N)^\top$, where $\mathbf{X}_i$ denotes an image of modality $\mathcal{M}_i$ when $c_i = 1$ and is a zero matrix otherwise. Using the dispatch matrix $\mathbf{D}$ and by applying some constraints, we dispatch N input images $\widetilde{\mathbf{X}}$ (including missing modalities) to N output images $\mathbf{S} = (\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_N)^\top$ by

$$\mathbf{S} = \mathbf{D}^\star \cdot \widetilde{\mathbf{X}}. \tag{5}$$

Here, $\mathbf{D}^\star$ is a constrained dispatch matrix based on the original dispatch matrix $\mathbf{D}$. Each element $\mathbf{D}^\star_{i,j}$ in the matrix $\mathbf{D}^\star$ denotes the weight of the input image $\mathbf{X}_j$ contributing to the output image $\mathbf{S}_i$.

We enforce constraints on $\mathbf{D}$ to ensure that: 1) self-dispatch based on the available input modality and 2) avoidance of

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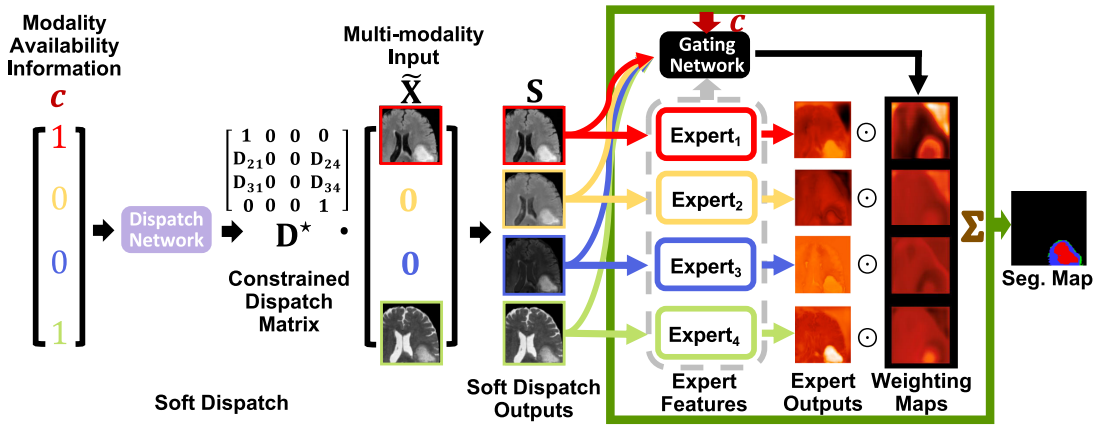


Fig. 2. Illustration of the soft dispatch strategy for MoME+ to handle heterogeneous multi-modality inputs in brain lesion segmentation. The modality availability information c is fed into the dispatch network, which generates a constrained dispatch matrix $\mathbf{D}^{\star}$. This matrix determines the weights for combining the input images $\widetilde{\mathbf{X}}$ and generating multi-channel images $\mathbf{S}$, termed as soft dispatch outputs. $\mathbf{S}$ serves as the input to the expert networks and c is treated as prior information embedded into the bottleneck of the gating network. The subsequent workflow aligns with the original MoME framework.

routing issues associated with a missing modality. This results in the constrained matrix $\mathbf{D}^{\star}$, formulated as

$$\mathbf{D}^{\star}_{i,j} = \begin{cases} \delta_{i,j} & \text{if } c_i = 1, \\ 0 & \text{if } c_j = 0, \\ \mathbf{D}_{i,j} & \text{else} \end{cases} \tag{6}$$

Here, $\delta_{i,j}$ is the Kronecker delta. When $c_i = 1$, $\mathbf{D}^{\star}$ enforces self-dispatch, and $\mathbf{S}_i = \mathbf{X}_i$; when $c_j = 0$, a missing modality $\mathbf{X}_j$ should not contribute to $\mathbf{S}_i$; else (i.e., $c_i = 0$ & $c_j = 1$) , an available modality $\mathbf{X}_j$ contributes to $\mathbf{S}_i$ based on the learned dispatch weight $\mathbf{D}_{i,j}$, which is a soft value between 0 and 1. An illustration of the dispatch process, using an example of $N = 4$ and $\boldsymbol{c} = (1, 0, 0, 1)$, can be found in Fig. 2. After the dispatch, there are no longer zero matrices in the output images $\mathbf{S}$. $\mathbf{S}$ now becomes the inputs to the expert networks, with each image $\mathbf{S}_i$ being fed to the corresponding expert E_i.

When training the MoME+ model including the dispatch network D, we randomly drop modalities, with each modality having an equal probability of being available or missing, resulting in $2^N - 1$ modality combinations. In this way, the dispatch network D learns to leverage the available modalities and perform a weighted combination of them to compensate for the missing modalities [55]. Similar to MoME, MoME+ uses the gating network to ensemble the expert outputs and the curriculum learning strategy, as described in Sec. III-B2 and Sec. III-B3. A difference is that the loss function Eq. (2) no longer involves only a single input image but multiple input images. The loss function $\mathcal{L}_{\text{cl}}^{+}$ is re-defined as

$$\mathcal{L}_{\text{cl}}^{+} = f_{\text{epoch}} \cdot \frac{1}{\|\boldsymbol{c}\|_1} \cdot \sum_{i=1}^{N} \mathbb{1}_{c_i=1} \cdot \mathcal{L}_i + (1 - f_{\text{epoch}}) \cdot \mathcal{L}_{\text{MoME}}. \tag{7}$$

Additionally, the modality available information $\boldsymbol{c}$ is treated as prior information embedded into the bottleneck of the gating network. Specifically, before embedding, we expand $\boldsymbol{c}$ into a one-hot tensor of shape $\mathbb{R}^{N \times d^3}$, where d^3 matches the shape of the 3D feature map at the bottleneck layer of the gating network.

D. Implementation Details

We implement each expert network and the gating network using the nnU-Net architecture [54], which is a strong segmentation backbone as shown in [22] and [54]. Additionally, all networks employ deep supervision [54]. The dispatch network for the multi-input MoME+ model utilises three layers of fully connected layers to achieve its lightweight task, where the numbers of channels are 4, 8, and 16 sequentially, and the activation functions are all ReLU [59]. As each expert network is expected to specialise in one imaging modality, the expert networks are first trained separately using a combined soft Dice and cross-entropy loss function to acquire the modality knowledge. Subsequently, the gating network is trained with the expert networks being fine-tuned using the curriculum learning loss in Eq. (2) for single modality input and Eq. (7) for heterogeneous multi-modality input respectively. All the hyperparameters (including $\{l_l\}_{l=1}^{L}$) follow the default setting in nnU-Net [54], except that 1) a lower learning rate of $1e^{-3}$ is set for the dynamic curriculum learning loss [38] and 2) the number of training epochs is set to 800, a sufficient number for convergence.

IV. Results

A. Dataset and Task Description

We curated six publicly available brain lesion segmentation datasets (MSSEG [60], OASIS [61], ISLES2022 [62], BraTS2021 [63], WMH2017 [33], and ATLAS2.0 [64]) and three in-house datasets (Tumour1,[1] Tumour2[1], and WMHsMix[2]), amounting to a total number of 6,585 3D images with brain lesion annotations. The dataset detail is reported in Table I. All images were rigidly registered to the MNI152 template using ANTs [65]. Brain masks were applied to the images for skull removal. Cropping was performed so that the

[1]Provided by Beijing Tiantan Hospital with ethical approval reference number: KY2022-078-04. Written informed consent was obtained from each participant according to the Declaration of Helsinki.

[2]Provided by Beijing Tiantan Hospital with ethical approval reference number: KY2021-028-03. Written informed consent was obtained from each participant according to the Declaration of Helsinki.

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TABLE I
AN OVERVIEW OF THE PUBLIC AND IN-HOUSE DATASETS USED FOR BRAIN LESION SEGMENTATION. NOTE THAT THE NUMBER OF AVAILABLE MODALITIES AND THEIR COMBINATIONS VARY ACROSS DATASETS

Datasets	Public datasets						In-house datasets		
	ATLAS2.0	ISLES2022	BraTS2021	WMH2017	MSSEG	OASIS	Tumour1	Tumour2	WMHsMix
Lesion type	Chronic stroke	Ischemic stroke	Glioma tumour	WMH	MS	Normal healthy	Glioma tumour	Glioma tumour	WMH Mixture[3]: PD, NMOSD, MS, SVD, AD
Modality	T1w	DWI	T1w, T1ce[1], T2w, FLAIR	T1w, FLAIR	T1w, T1ce, T2w, FLAIR	T1w, FLAIR[2]	T2w	FLAIR	FLAIR
#Training images	545	150	4,168	60	36	200	29[4]	27[4]	20[4]
#Test images	110	100	836	60	24	100	40	40	40

[1] Annotations for tumour core.
[2] In-house expert annotations based on FLAIR.
[3] A dataset of mixed disease types, including the Parkinson's Disease (PD), Neuromyelitis Optica Spectrum Disorder (NMOSD), Multiple Sclerosis (MS), Small Vessel Disease (SVD), and Alzheimer's Disease (AD).
[4] Not used for foundation model training, but used for training task-specific models regarded as upperbound for generalisation onto unseen datasets.

size of all images was standardised to $160 \times 196 \times 160$ with a voxel spacing of $1 \times 1 \times 1$ mm^3. Each of the six public datasets was split into training and test sets, as detailed in Table I. The MoME model was trained on the combined training sets from all six public datasets. It was then evaluated on the test sets from the public datasets (**seen datasets**) and the three in-house datasets (**unseen datasets**), which were never used for training and allowed the assessment of the generalisation of MoME. For assessing the extended multi-input MoME+ model, we used the BraTS2021 dataset, which contains T1w, T1ce, T2w, and FLAIR modalities, and simulated different scenarios of missing modalities. Specifically, for the four modalities, each can either be available or unavailable, with the constraint that at least one modality must be present. This results in a total of $2^4 - 1$ possible combinations, and their details are presented in Table VII.

B. Competing Task-Specific and Foundation Models

MoME was compared with the following task-specific and foundation models for single modality input settings.

Task-specific nnU-Nets: The nine datasets cover a variety of brain lesions and modalities. We regard the segmentation of each modality for each dataset as a distinct task [24], resulting in a total number of 17 tasks (14 tasks for seen datasets and 3 tasks for unseen datasets). Thus, 17 supervised task-specific nnU-Net models were trained.

nnU-Net: A single nnU-Net [54] was trained with the combined training sets from all public datasets. It can be viewed as a baseline foundation model.

Multi-Talent [22]: Multi-Talent demonstrates superior performance in handling conflicting class definitions for multi-dataset training and shows promising lesion segmentation results. We trained it with the combined training sets.

Hermes [23]: Hermes is a foundation model that injects context prior knowledge (e.g. anatomical region and imaging modality) into the segmentation network to address the challenges of data and modality diversities. We adapted Hermes for our task and incorporated the imaging modality information as prior knowledge. Hermes was also trained with the combined training sets.

nnU-Net ResEnc [66]: nnU-Net ResEnc employs a residual encoder U-Net within the nnU-Net framework. It is optimised for various hardware configurations, including nnU-Net ResEnc M, L, and XL. As [66] highlights the importance of scalability for larger datasets, we chose the advanced nnU-Net ResEnc XL and trained with the combined training sets.

SAM-Med3D [17]: This is a SAM-based [16] model trained on a large-scale 3D medical image dataset. We fine-tuned SAM-Med3D with the combined training sets and provided a 10-point prompt [17] during the inference.

For heterogeneous multi-modality input, the competing foundational models cannot be applied. We compared the MoME+ model with $2^N - 1$ dedicated nnU-Nets, each trained for one combination of modalities. For the BraTS2021 dataset, $N = 4$ modalities were involved.

All models were trained for 800 epochs on Nvidia A100 GPUs. We used the default hyperparameters for the transformer-based SAM-Med3D. All the other models were based on nnU-Net and used its default hyperparameters.

C. Performance of MoME on Seen Datasets

Table II compares the average Dice score and Hausdorff distance (HD) (mm) of MoME with the competing models and reports the overall GPU memory usage and GPU hours for training each model. To mitigate the bias caused by the varying numbers of images in different test sets, we report the Dice score with different levels of averaging: across the 14 seen tasks (task-level), across 6 seen datasets (dataset-level), and across 1,230 test images (image-level). MoME achieves better performance compared to other foundation models in terms of both Dice score and HD, for example, achieving 2% to 4% higher Dice score than nnU-Net, Multi-Talent, Hermes, and nnU-Net ResEnc XL. It outperforms SAM-Med3D by a large margin of 7% to 15%. This is possibly because SAM-Med3D requires point prompts to indicate a foreground region, which may not be suitable for brain lesion segmentation, where lesions can be numerous, small, and scattered across the brain, such as the WMH lesions [5], [33]. The performance of MoME is comparable to task-specific nnU-Nets. However, the latter requires the training of 14 separate models, and its total amount of GPU memory consumption is about three times that of MoME.

A more detailed analysis of the results is shown in Fig. 3. Fig. 3a presents a comprehensive comparison between the foundation models from eight perspectives, reporting the average Dice score for five different modalities and three

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TABLE II
AVERAGE DICE SCORE AND HAUSDORFF DISTANCE (mm) OF MOME AND THE COMPETING METHODS ON SEEN DATASETS. THE BEST RESULTS AMONG THE FOUNDATION MODELS ARE HIGHLIGHTED IN BOLD. THE GPU MEMORY USAGE AND GPU HOURS FOR MODEL TRAINING ARE ALSO REPORTED

Dice | HD (mm)	Foundation Model						Task-specific nnU-Nets
	nnU-Net	SAM-Med3D	Multi-Talent	Hermes	nnU-Net ResEnc XL	MoME	
Image-level	0.8007 | 28.74	0.7322 | 26.83	0.7948 | 26.86	0.8017 | 30.54	0.8084 | 25.22	**0.8204** | **25.18**	0.8202 | 24.21
Task-level	0.6654 | 38.37	0.5249 | 40.80	0.6825 | 37.08	0.6720 | 38.28	0.6838 | 37.13	**0.7026** | **35.27**	0.7099 | 32.05
Dataset-level	0.6561 | 39.30	0.5083 | 43.08	0.6681 | 38.93	0.6603 | 40.16	0.6718 | 38.41	**0.6938** | **35.90**	0.6950 | 34.50
GPU memory usage (GB)	7.5	32.5	11.9	7.5	48.3	38.2	7.5 (×14) [1]
GPU hours	9.1	50.1	18.8	10.2	41.2	46.6	9.1 (×14) [2]

[1] Fourteen task-specific nnU-Net models, each using around 7.5 GB of GPU memory.
[2] Fourteen task-specific nnU-Net models, each using around 9.1 GPU hours.

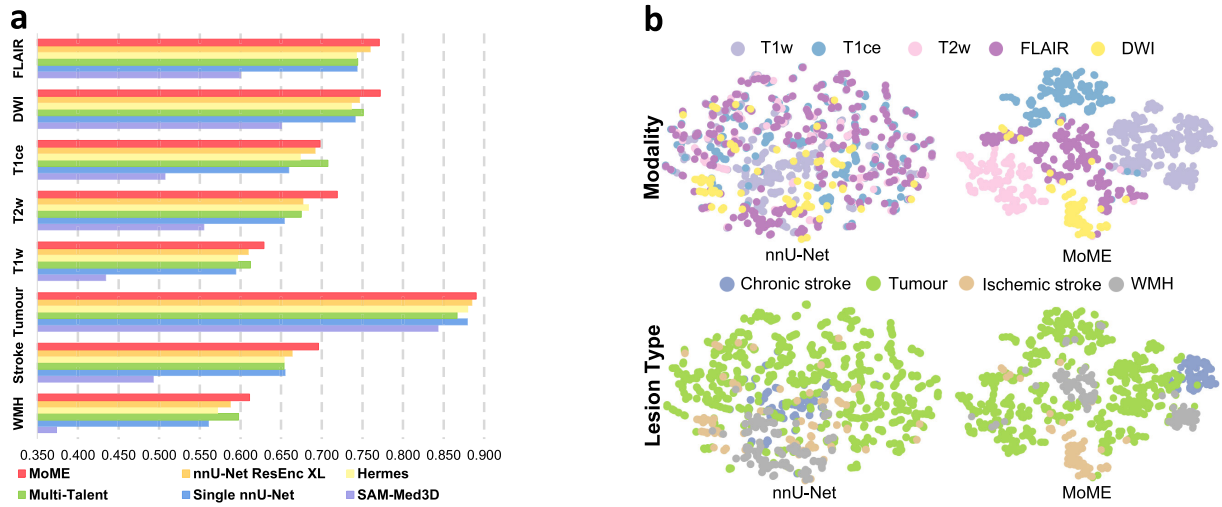


Fig. 3. More detailed analysis of the MoME result on seen datasets. a) A bar chart that compares the average Dice score of foundation models from the perspectives of different modalities and lesion types. b) t-SNE plots of latent spaces for the single nnU-Net (trained on all imaging modalities) and MoME, where each dot represents a brain image.

different lesion types. For seven out of the eight perspectives, MoME outperforms the other foundation models. Among the modalities, the Dice metric is lowest for T1w images, likely because detecting lesions is more challenging on T1w, where lesions do not manifest as clearly as on FLAIR and T2w. Fig. 3b analyses the latent features obtained at the bottleneck of the gating network in MoME and the single nnU-Net using t-SNE plots [67]. It shows that MoME achieves a more discriminative clustering for images of different modalities and lesion types compared to the single nnU-Net, which may be relevant to its improved performance.

D. Generalisation of MoME to Unseen Datasets

Table IV reports the segmentation performance on the three unseen in-house datasets for evaluating model generalisability. Multi-Talent is not included as it cannot be directly applied to unseen datasets. Again, MoME outperforms other competing foundation models. On two unseen datasets (Tumour1 and Tumour2), it even exceeds the task-specific nnU-Nets, which can be considered the upper-bound benchmark as they had access to the training data from the unseen datasets. On the other unseen dataset (WMHsMix), MoME still performs relatively well and achieves an average Dice score of 0.7015, where the dataset contains diverse WMH lesions of five diseases (Table I).

TABLE III
AVERAGE DICE SCORE AND HAUSDORFF DISTANCE (MM) ON THREE UNSEEN DATASETS. THE TOP SCORES ARE IN BOLD. ASTERISKS INDICATE THAT THE DIFFERENCE BETWEEN THE PROPOSED METHOD AND THE COMPETING METHOD IS STATISTICALLY SIGNIFICANT (*: $p \leq 0.05$, **: $p \leq 0.01$, ***: $p \leq 0.001$) USING A PAIRED STUDENT'S t-TEST

Datasets	Foundation Model					Task-specific nnU-Nets
	nnU-Net	SAM-Med3D	Hermes	nnU-Net ResEnc XL	MoME	
Tumour1	0.8358***	0.8339	0.8446*	0.8373**	**0.8545**	0.8518
Tumour2	0.7959	0.8146***	0.7968**	0.8138	**0.8293**	0.8157
WMHsMix	0.7032	0.5479***	0.6968	0.6898	0.7015	**0.7508****
Dice	0.7783	0.7322	0.7794	0.7803	0.7951	**0.8061**
HD (mm)	27.95	24.83	28.66	25.27	23.98	**23.24**

E. Summary of MoME Results on Both Seen and Unseen Datasets

Table IV reports the detailed results on both seen and unseen datasets, highlighting the performance of the proposed MoME method across 17 tasks (identified by IDs ranging from 1 to 17), nine datasets, three major types of brain lesions, and five modalities. The majority of these results not only outperforms competing methods but also demonstrates performance comparable to task-specific nnU-Nets. Importantly, MoME seamlessly handles any input modality and lesion type, without requiring retraining when new datasets are introduced, distinguishing it from task-specific nnU-Nets.

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TABLE IV

DETAILED AVERAGE DICE SCORE OF MoME AND COMPETING METHODS FOR EACH TASK, LESION TYPE, AND MODALITY. THE TOP RESULTS, EXCLUDING THE TASK-SPECIFIC nnU-Nets RESULTS, ARE HIGHLIGHTED IN BOLD

Settings	Task			nnU-Net	SAM-Med3D	Multi-Talent	Hermes	nnU-Net ResEnc XL	MoME	Task-Specific nnU-Nets
	Dataset	Modality	Task ID							
Seen datasets	BraTS2021	T1w	1	0.8344	0.7908	0.8192	0.8350	0.8400	**0.8510**	0.8573
		T2w	2	0.8868	0.8599	0.8690	0.8893	0.8930	**0.8995**	0.8981
		T1ce	3	0.8816	0.8271	0.8708	0.8825	**0.8878**	0.8856	0.8888
		FLAIR	4	0.9165	0.8956	0.9078	0.9146	0.9186	**0.9220**	0.9229
	ATLAS2.0	T1w	5	0.5685	0.3354	0.5561	0.5719	0.5808	**0.6203**	0.5645
	WMH2017	FLAIR	6	0.6726	0.5124	0.6681	0.6723	**0.6951**	0.6909	0.7341
		T1w	7	0.5387	0.3684	0.5528	0.5511	0.5623	**0.5773**	0.5677
	OASIS	T1w	8	0.5638	0.4354	0.5932	0.5626	0.5742	**0.5937**	0.6028
		FLAIR	9	0.7241	0.5753	0.7431	0.7261	0.7350	**0.7737**	0.7970
	ISLES2022	DWI	10	0.7415	0.6510	0.7510	0.7368	0.7471	**0.7723**	0.7824
	MSSEG	T1w	11	0.4664	0.2426	**0.5408**	0.4630	0.4913	0.5044	0.5512
		T2w	12	0.4219	0.2501	0.4804	0.4781	0.4604	**0.5384**	0.4933
		T1ce	13	0.4375	0.1874	**0.5445**	0.4646	0.4971	0.5099	0.5586
		FLAIR	14	0.6619	0.4176	0.6584	0.6600	0.6907	**0.6979**	0.7199
Unseen datasets	Tumour1	FLAIR	16	0.8358	0.8339	-	0.8446	0.8373	**0.8545**	0.8518
	Tumour2	T2w	15	0.7959	0.8146	-	0.7968	0.8138	**0.8293**	0.8157
	WMHsMix	FLAIR	17	**0.7032**	0.5479	-	0.6968	0.6898	0.7015	0.7508
Lesion level	Tumour		1,2,3,4 15,16	0.8710	0.8370	-	0.8732	0.8753	**0.8825**	0.8838
	WMH		6,7,8,9,10, 11,12,13, 14,17	0.5767	0.3930	-	0.5861	0.5995	**0.6208**	0.6417
	Stroke		5,10	0.6550	0.4932	0.6535	0.6544	0.6640	**0.6963**	0.6734
Modality Level	T1w		1,5,7,8,11	0.6082	0.4511	0.6273	0.6081	0.6216	**0.6424**	0.6440
	T2w		2,12,15	0.7015	0.6416	-	0.7214	0.7224	**0.7557**	0.7357
	T1ce		3,13	0.6595	0.5072	**0.7076**	0.6736	0.6925	0.6977	0.7237
	FLAIR		4,6,9, 14,16,17	0.7523	0.6305	-	0.7524	0.7611	**0.7734**	0.7961

F. Ablation Studies

Ablation studies were conducted on both seen and unseen datasets to verify the contributions of the designs in MoME with results presented in Table V. For the modality experts, ✘/✔ denotes randomly initialised experts versus specialised experts based on the imaging modality; for the hierarchical gating, ✘/✔ denotes the basic gating without hierarchical structure versus our hierarchical gating; and for the curriculum learning, ✘/✔ denotes the application of Eq. (4) versus Eq. (2) during model training. When the three contributions are sequentially removed, even though the number of model parameters remains constant in this ablation study, the segmentation performance tends to decrease gradually, confirming the benefits of these contributions. This indicates that the improvement in MoME is not solely due to the larger model size, but also due to the modality-based expert design and the training strategy, which better balances expert collaboration and specialisation within the MoME framework.

TABLE V

ABLATION STUDIES OF MoME. THE BEST SCORES ARE IN BOLD

Modality experts		✔	✔	✔	✘
Hierarchical gating		✔	✔	✘	✘
Curriculum learning		✔	✘	✘	✘
Seen Datasets	Image-level	**0.8204**	0.8142	0.8117	0.8027
	Task-level	**0.7026**	0.6969	0.6878	0.6529
	Dataset-level	**0.6938**	0.6852	0.6777	0.6488
Unseen Datasets		**0.7951**	0.7889	0.7912	0.7814

G. Expert Degeneration

We investigated the issue of expert degeneration by comparing the expert output, weighted by gating, without and with the proposed curriculum learning strategy. Detailed results can be found in Fig. 4. In Fig. 4a, without an anti-degeneration strategy, T2w and FLAIR experts lose their acquired modality knowledge, i.e. the sensitivity to the brain lesion on the corresponding modalities. In contrast, Fig. 4b shows that with the anti-degeneration strategy, expert degeneration is effectively avoided. Each expert is activated according to its specialised modality or related modalities, as indicated by the blue dashed boxes. In particular, we observe that some modality experts exhibit a strong relationship and can be activated in tandem, such as T1w and T1ce, which are consistently activated together, aligning with our understanding of their relatedness in brain imaging. When activated in tandem, these modality experts provide complementary information, which, combined with the gating aggregation, helps to refine the segmentation performance. This comparison underscores the risk of expert inactivity and loss of modality-specific expertise in the absence of the curriculum learning anti-degeneration strategy. Conversely, our proposed approach ensures that experts maintain responsiveness to modality-specific information.

H. Improving the Model Efficiency

We conducted additional experiments to explore the potential for improving model efficiency through the impact of weight sharing among modality experts, with a specific focus on the decoder. In our original setup, as shown in Fig. 3b, the modality-specific encoder allows each expert to capture distinct features unique to each modality, which

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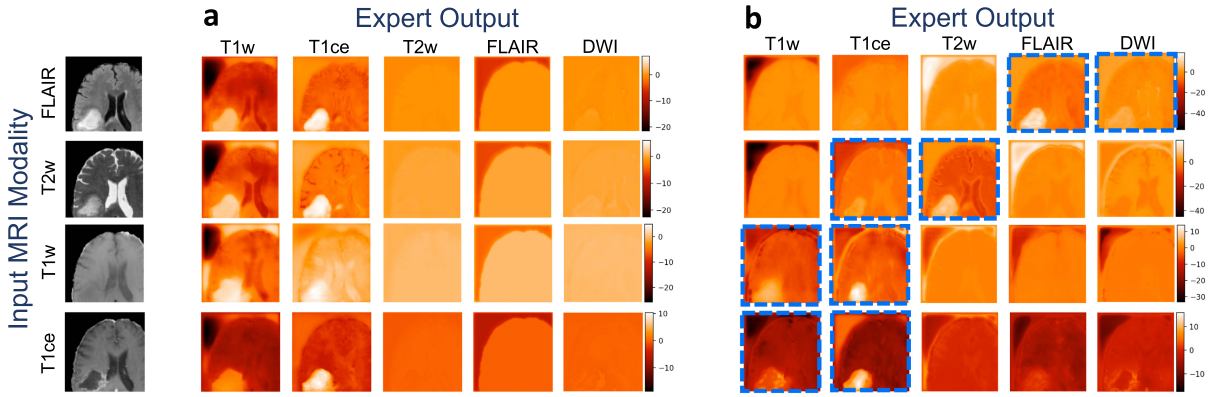


Fig. 4. The output of each modality expert weighted by the gating mechanism achieved a) without or b) with the proposed anti-degeneration curriculum learning strategy. The blue dashed boxes show the brain lesion highlighted by the expert outputs.

TABLE VI
DICE SCORE AND MODEL EFFICIENCY COMPARISONS FOR THE SINGLE nnU-Net, MoME WITH A SHARED DECODER, AND THE ORIGINAL MoME. THE HIGHEST DICE SCORE ARE MARKED IN BOLD

Models	Seen dataset	Unseen datasets	GPU memory usage (GB)	Model storage (GB)
Single nnU-Net	0.8007	0.7783	7.5	0.33
MoME (shared decoder)	0.8133	0.7949	32.7	1.19
MoME	**0.8204**	**0.7951**	38.2	1.98

we found improved segmentation performance for modality-specific characteristics. Based on that, we decided to create an alternative configuration where all modality experts share the same decoder.

To evaluate the shared-decoder configuration, we compared three models: (1) single nnU-Net, (2) MoME with a shared decoder, and (3) the original MoME. Table VI summarises these results, comparing segmentation performance on the average image level for both seen and unseen datasets, along with GPU memory usage and model storage.

When comparing MoME (shared decoder) to single nnU-Net, the improvement in Dice performance highlights the benefits of a specialised encoder. Further comparing the MoME (shared decoder) to the original MoME, our findings reveal that while the shared decoder version achieves comparable generalisation performance on unseen datasets, it exhibits a slight decline in segmentation accuracy on seen datasets. This emphasises the importance and advantages of incorporating a specialised decoder for further enhancements.

On the other hand, the MoME (shared decoder) is computationally more efficient compared to the original MoME, in terms of both the GPU memory usage and model storage size. GPU memory usage is reduced by 14% and the number of parameters required for model storage is reduced by approximately 40%, compared to the original MoME.

I. Extension to Multi-Modality Input

Table VII presents the average Dice performance of the extended MoME+ model for multiple input images with missing modalities, compared to 2^4-1 dedicated nnU-Nets trained with different modality combinations on the BraTS2021 dataset. Each dedicated nnU-Net was trained with a specific modality combination, as detailed in Table VII. The other

TABLE VII
AVERAGED DICE SCORE WITH THE STANDARD DEVIATION OF MoME+ AND 2^4-1 DEDICATED nnU-Nets FOR 3-CLASS SEGMENTATION ON THE BraTS2021 DATASET. THE DETAILED INPUT MODALITY COMBINATION USED FOR TRAINING EACH DEDICATED nnU-Net IS LISTED. • MEANS THE CORRESPONDING MODALITY IS AVAILABLE, AND ○ MEANS IT IS NOT

Method	#Models	Mean	Whole Tumour	Tumour Core	Enhancing Tumour
Dedicated nnU-Nets	2^4-1	$0.8366_{\pm 0.12}$	$0.9061_{\pm 0.07}$	$0.8456_{\pm 0.19}$	$0.7583_{\pm 0.18}$
MoME+	1	$0.8334_{\pm 0.13}$	$0.9046_{\pm 0.08}$	$0.8422_{\pm 0.19}$	$0.7534_{\pm 0.18}$

2^4-1 Modality Combinations

Modality	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
T1w	•	○	○	○	•	•	•	○	○	○	•	•	•	○	•
T1ce	○	•	○	○	•	○	○	•	•	○	•	•	○	•	•
T2w	○	○	•	○	○	•	○	•	○	•	•	○	•	•	•
FLAIR	○	○	○	•	○	○	•	○	•	•	○	•	•	•	•

foundational models, Hermes, SAM-Med3D, Multi-Talent and nnU-Net ResEnc XL, were not included in the experimental results since they cannot handle heterogeneous input modalities. Table VII demonstrates that a single MoME+ model achieves performance comparable to 2^4-1 dedicated nnU-Nets, with only 0.32% drop in the average Dice score across 2^4-1 modality combination scenarios. Furthermore, the comparable standard deviation relative to the 2^4-1 dedicated modality-specific nnU-Net models indicates that MoME+ delivers consistent and reliable performance. But the latter requires significantly more model storage and training resources. Specifically, in terms of GPU usage for training, MoME+ requires 30.56GB, while 2^4-1 modality-specific dedicated nnU-Nets demand over 112.5GB. This represents a reduction of more than 72.8% in GPU usage compared to the dedicated nnU-Nets for handling heterogeneous input modalities. In terms of model storage, MoME+ requires approximately 1.65GB, while 2^4-1 modality-specific dedicated nnU-Nets require 4.95GB. This indicates a more than 66.6% reduction in storage.

V. DISCUSSION

Foundational models have gained popularity and developed rapidly in the natural image domain. However, they do not account for the complexity of medical imaging modalities. In this paper, MoME and MoME+ have been proposed with the aim of creating a task-agnostic model capable of

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handling various modalities and lesion types in clinical brain MRI, regardless of the modality or modality combinations. The diversity of imaging modalities is unique to the field of medical imaging, presenting challenges in developing a unified foundational model. However, this complexity also offers a unique opportunity to advance foundational model development by exploiting the complementary relationships between modalities. This motivates us to choose the Mixture of Experts (MoE) architecture and regard each modality as an expert within this framework.

MoME demonstrates its clinical potential by serving as a unified foundation model for brain lesion segmentation, achieving comparable performance to numerous task-specific nnU-Nets. This approach enhances user-friendliness for clinicians by eliminating the need to train and manage multiple dedicated models tailored to a specific modality and lesion type. In practical settings, the variability of patient imaging modalities and lesion types makes managing numerous task-specific models impractical and burdensome within a hospital's neuroimaging department. Adopting a unified foundation model like MoME simplifies this complexity, thereby improving clinical workflow efficiency. Furthermore, we extend MoME from a single input scenario to MoME+ to address the multi-input scenario, with potential missing modalities and heterogeneous combinations of input modalities. This is a challenge not typically addressed by existing foundation models designed for natural images. Through innovations like the soft dispatch network and embedding modality information into the gating network, we demonstrate the capability of MoME+ to manage diverse inputs with the potential to handle more complex scenarios in clinical neuroimaging. Additionally, although MoME is specifically designed for brain lesion segmentation, it is feasible to extend this approach to a wider range of applications. For instance, it could be applied to organ segmentation tasks or other domains that involve multiple modality inputs, where the collaboration between modality-based experts may benefit the task performance.

At the moment, most publicly available and in-house datasets annotate only one type of lesion per brain image. However, for some patients, more than one lesion types can exist in the same brain image, and not all of these lesion types are annotated in the current datasets. Improving the dataset annotation accuracy could be achieved in future work by implementing quality control measures or employing advanced data engines [16]. Furthermore, we can generate additional training samples using lesion-aware data augmentation techniques such as CarveMix [2], [68]. This method can generate augmented brain images mixed with multiple types of lesions, along with augmented annotations. This will enrich the training datasets to encompass more diverse variations of brain lesion types, which can further improve the foundation model for brain lesion segmentation.

VI. Conclusion

We have developed a novel universal foundation model MoME for brain lesion segmentation on MRI, where a mixture of modality experts is incorporated to address the challenge of modality diversity. MoME enables collaboration between experts with hierarchical gating and mitigates expert degeneration with a curriculum learning strategy. Additionally, we have extended MoME to MoME+ with soft routing dispatch, allowing it to handle heterogeneous multi-modality inputs. Extensive experimental results on nine public and in-house datasets and 17 tasks show that it outperforms competing foundation models across different modalities and lesion types. It is also more efficient than task-specific models, providing a flexible way for model deployment.

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